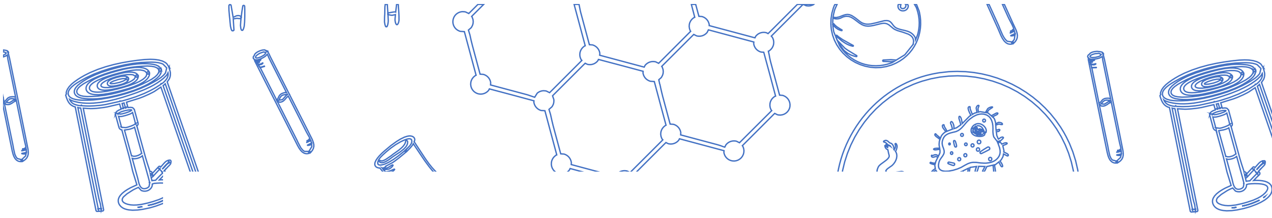
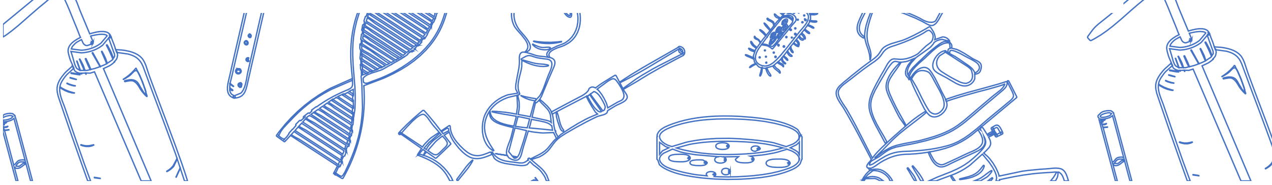


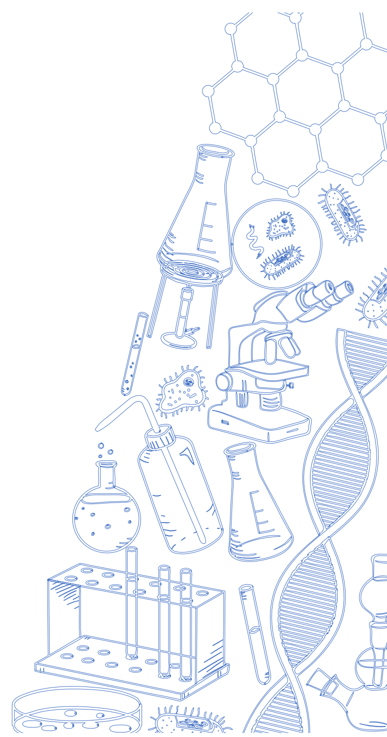
Science G10

Chemical Reactions
Class 2



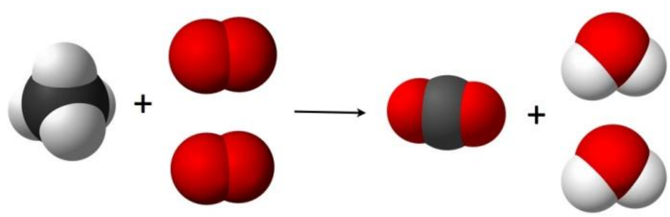
Agenda

1. Quiz 2
2. Lesson 2
   - Law of Conservation of Mass
   - Balancing Chemical Reactions
   - Types of Chemical Reactions
     - Synthesis
     - Decomposition
     - Combustion
     - Single Displacement
     - Double Displacement



Law of Conservation of Mass

- In any given reaction, the total mass of the reactants equals the total mass of the products
- Atoms cannot be created or destroyed



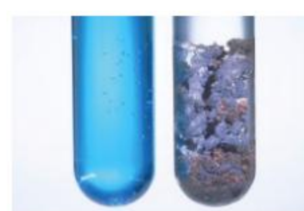
$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

Chemical Reactions

REACTANTS		PRODUCTS

$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$

Reactants	Products
1 Zn	1 Zn
1 Cu	1 Cu
1 (SO_4)	1 (SO_4)



State Symbols

- State symbols are often written behind the chemical formula to indicate the state of the substance

$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$$

State Symbol	Meaning
(s)	Solid
(l)	Liquid/Molten
(g)	Gaseous
(aq)	Aqueous (dissolved in water)

Balancing Chemical Reactions

Word Equation	Iron reacts with oxygen to form iron(III) oxide
Skeleton Equation	$Fe + O_2 \rightarrow Fe_2O_3$
Number of Atoms	1 Fe 2 O → 2 Fe 3 O
Balanced Reaction (add coefficients)	$\mathbf{4}\ Fe + \mathbf{3}\ O_2 \rightarrow \mathbf{2}\ Fe_2O_3$

General Rules for Balancing Chemical Reactions:

- Only add coefficients, do not add subscripts
- Balance non-hydrogens and non-oxygens first
- Balance polyatomic ions as a unit
- All coefficients in the balanced chemical reaction need to be in simplest form as positive integers
- If chemical reaction cannot be balanced, check if the crisscrossing was done correctly

STOP Checkpoint STOP

Balance the following chemical reactions:

a) $\_\_K_2O \rightarrow \_\_K + \_\_O_2$

b) $\_\_KClO_3 \rightarrow \_\_KCl + \_\_O_2$

c) $\_\_AlCl_3 + \_\_Na_2CO_3 \rightarrow \_\_Al_2(CO_3)_3 + \_\_NaCl$

d) $\_\_NaOH + \_\_H_2SO_4 \rightarrow \_\_Na_2SO_4 + \_\_H_2O$

e) $\_\_N_2O_5 \rightarrow \_\_N_2O_4 + \_\_O_2$

Checkpoint

Write the balanced chemical reaction of:

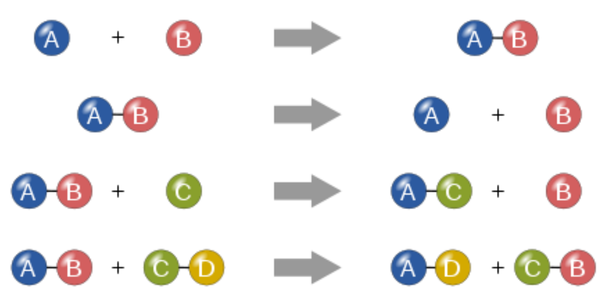
a) Magnesium with oxygen

b) Methane (CH_4) burns in oxygen to produce carbon dioxide and water

c) Zinc metal reacts in silver nitrate solution to produce zinc nitrate and silver metal

Types of Chemical Reactions

- Synthesis
- Decomposition
- Combustion
- Single Displacement
- Double Displacement
  - Neutralization



Synthesis

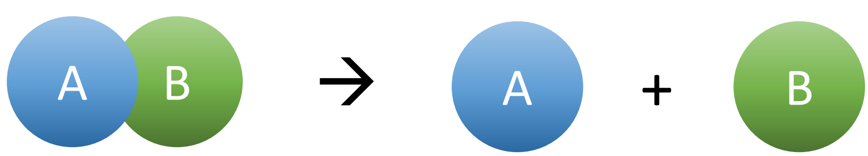
Examples:

$$3H_2 + N_2 \rightarrow 2NH_3$$

$$MgO + CO_2 \rightarrow MgCO_3$$

$$2CO + O_2 \rightarrow 2CO_2$$

Decomposition



Examples:

$$2H_2O \rightarrow 2H_2 + O_2$$

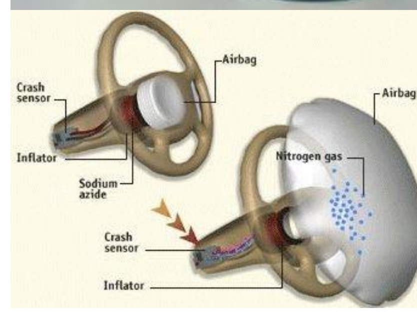
$$2Fe(OH)_3 \rightarrow Fe_2O_3 + 3H_2O$$

$$H_2CO_3 \rightarrow CO_2 + H_2O$$

Decomposition of Sodium Azide

$$2NaN_3(s) \rightarrow 2Na(s) + 3N_2(g)$$

- Airbags contain sodium azide, $NaN_3(s)$
- During a collision, electrical energy triggers the rapid decomposition of $NaN_3(s)$ into nitrogen gas to inflate the airbag



Checkpoint

Write the balanced chemical equation for the following and classify the type of reaction:

a) Zinc chloride → zinc + chlorine

b) Potassium oxide + water → potassium hydroxide

Combustion

- **Combustion** is a reaction in which a substance reacts with oxygen gas, releasing energy in the form of light and heat
- Two types:
  - **Complete combustion** occurs when there is sufficient oxygen
  - **Incomplete combustion** occurs where there is insufficient oxygen

Complete Combustion of Hydrocarbons

- Hydrocarbons are compounds that contain carbons and hydrogens
- Complete combustion of hydrocarbons produces CO_2, H_2O, and energy
- Flame is blue in colour with little smoke

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + \text{energy}$$

Checkpoint

Assuming complete combustion, write the balanced chemical reaction for the combustion of:

a) Propane (C_3H_8)

b) Octane (C_8H_{18})

Incomplete Combustion of Hydrocarbons

- Incomplete combustion of hydrocarbons produces CO_2, CO, H_2O, C(s), and energy
- Flames from incomplete combustion tend to be yellow or orange in colour and are smokier due to unburned carbon

Air Hole Closed

$$C_4H_{10} + 5O_2 \rightarrow 2CO_2 + 5H_2O + CO + C(s) + \text{energy}$$

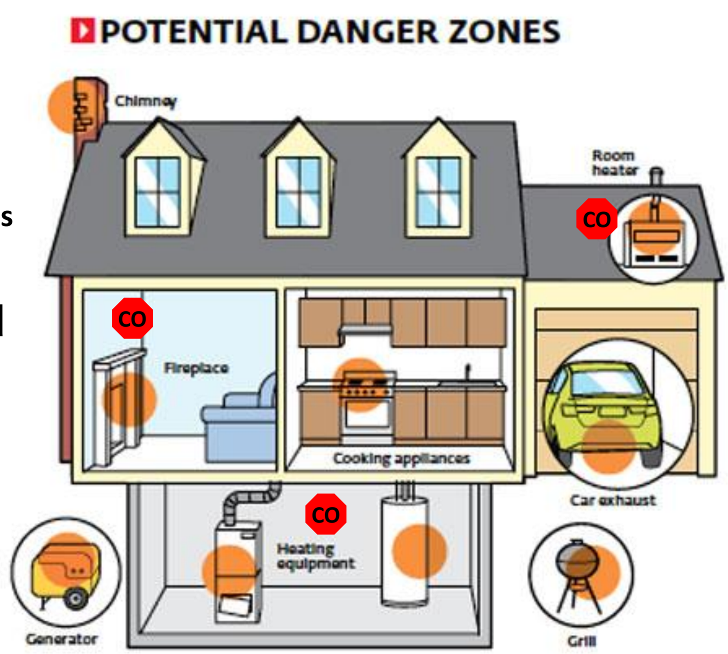
Dangers of Carbon Monoxide

- Carbon monoxide displaces oxygen in the blood and deprives the heart, brain, and other vital organs of oxygen
- Carbon monoxide is colourless, odourless, and tasteless

Symptoms of carbon monoxide poisoning:

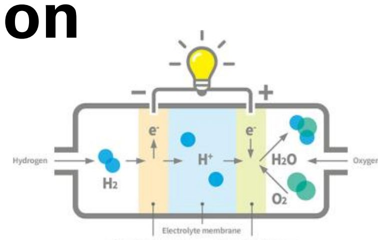
CO **Placement of CO(g) alarms**

- It is recommended that at least one alarm is installed on every level



- October 24, 1993, Robert Latimer killed his 13-year old daughter Tracy Latimer by piping the exhaust fumes containing CO(g) into the interior of the car where she was placed
- Tracy had a severe form of cerebral palsy and suffered considerable pain
  - Father killed her to relieve her of her pain
- Triggered debates around health ethics and euthanasia

Other Forms of Combustion



Combustion of Hydrogen (Fuel Cells)

$2H_2(g) + O_2(g) \rightarrow 2H_2O(g) + \text{energy}$

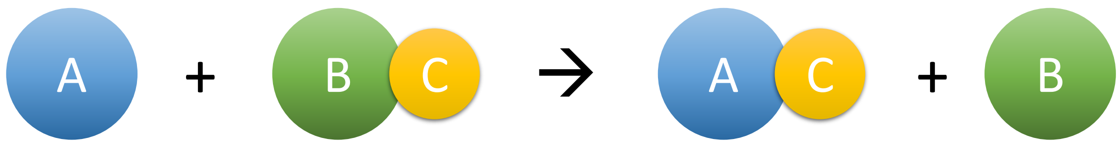
Combustion of Nitrogen-Containing Hydrocarbons

$C_xH_yN_z + O_2 \rightarrow CO_2(g) + H_2O(g) + N_2(g)$

Combustion of Phosphorus

$P_4(s) + 5O_2(g) \rightarrow P_4O_{10}(g) + \text{energy}$

Single Displacement



- Metal displaces the metal (use the most common ionic charge for multivalent metals unless indicated otherwise)

$$Cu(s) + 2AgNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2Ag(s)$$

- Nonmetal displaces the nonmetal

$$F_2(g) + 2KCl(aq) \rightarrow 2KF(aq) + Cl_2$$

Checkpoint

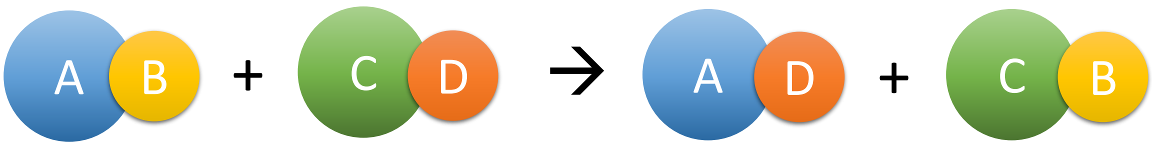
Predict the products of the following single displacement reactions and balance:

a) $Fe + Pb(NO_3)_2 \rightarrow$

b) $Cl_2 + NaI \rightarrow$

c) $Li + H_2O \rightarrow$

Double Displacement



- Metals trade places or nonmetals trade places

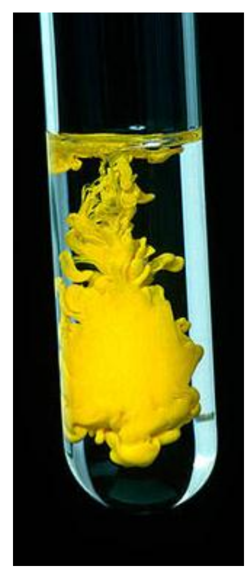
  $AgNO_3(aq) + NaCl(aq) \rightarrow NaNO_3(aq) + AgCl(s)$

- Double displacement reactions occur if:
  - A solid precipitate forms OR H_2O forms OR gas forms

Precipitate

- **Precipitate** is a solid formed from the reaction of two solutions; an insoluble compound

$Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow \mathbf{PbI_2(s)} + 2KNO_3(aq)$



$PbI_2(s)$ is a yellow precipitate

Checkpoint

Predict the products of the following double displacement reactions and balance:

a) $CuS + KCl \rightarrow$

b) $CaBr_2 + KOH \rightarrow$

c) $Ca(OH)_2 + H_3PO_4 \rightarrow$

Checkpoint

Write the balanced chemical equation for the following and classify the type of reaction:

a) Silver nitrate + sodium phosphate → silver phosphate precipitate + sodium nitrate

b) Calcium + water → hydrogen gas + calcium hydroxide

What I Learned Today:

- ❑ Law of Conservation of Mass
- ❑ Balancing Chemical Reactions
- ❑ Types of Chemical Reactions
  - Synthesis
  - Decomposition
  - Combustion (Complete and Incomplete)
  - Single Displacement
  - Double Displacement

Due next class: Class 2 Homework